



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

LAB REPORTS

Course no: **MTE 3206**

Course Title: **Hydraulic and Pneumatic Control Sessional**

Experiment No: **05**

Name of the Experiment: Design and Investigate the Stable and Unstable Control of a Double-Acting Cylinder.

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Experiment No.: 05

Experiment Name: Design and Investigate the Stable and Unstable Control of a Double-Acting Cylinder

Objectives

- To understand the principles governing electro-pneumatic control systems.
- To configure the DL 8115EPB trainer to simulate various stability conditions.
- To observe and analyse the double-acting cylinder's behaviour under these conditions.
- To identify factors contributing to stable and unstable control.

Theory

A double-acting hydraulic cylinder is a mechanical actuator that uses pressurized hydraulic fluid to generate linear motion and force during both extension and retraction. In contrast to single-acting cylinders, which exert force in only one direction and depend on external forces (such as springs or gravity) for the return stroke, double-acting cylinders apply hydraulic pressure to both sides of the piston, enabling controlled movement in both directions [1].

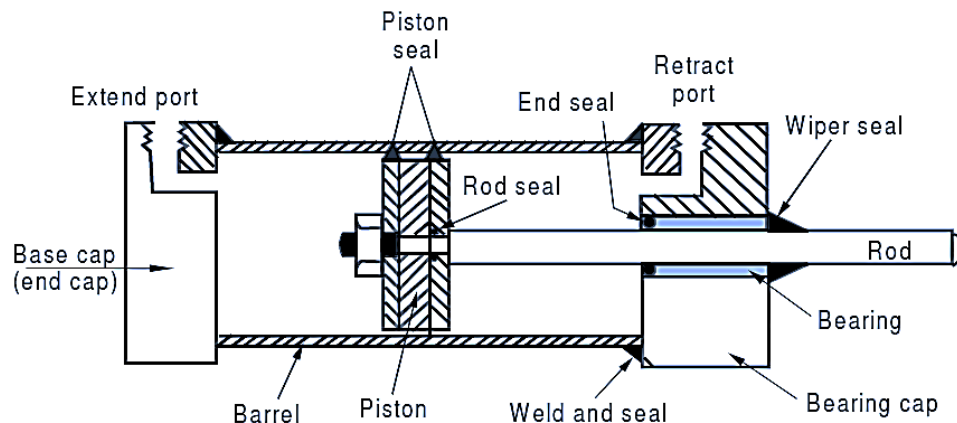


Fig.5.1. Double Acting Cylinder

Components

- **Cylinder Barrel:** A tube-like structure that houses the piston and hydraulic fluid.
- **Piston:** A disc inside the barrel that separates it into two chambers and moves when hydraulic pressure is applied.
- **Piston Rod:** A rod connected to the piston that extends through one end of the cylinder to transfer force to external components.
- **Seals:** Prevent hydraulic fluid leakage and help maintain internal pressure.
- **Hydraulic Ports:** Two ports positioned at either end of the cylinder for fluid entry and exit

Required Apparatus

- De Lorenzo Electro Pneumatic Machine (DL 8115EPB)
- Double-acting pneumatic cylinder
- Push buttons (switches)
- Regular Switches
- 3/2-way solenoid valve
- Pneumatic tubing and connectors
- Air compressor
- Power supply (24V DC)
- Electrical connecting wires

Circuit Buildup

Condition 01: Unstable Electro-drive with Unstable Control

Electro-drive: A push button is used to trigger the solenoid. Since the push button returns to its original state when released, the electro-drive remains unstable.

Control: A spring return is installed on one side of the directional control valve (DCV), ensuring the valve automatically returns to its default position when not actuated.

Setup:

- Connect the push button to the solenoid valve.
- Set up the DCV with a spring return on one side.
- Link the DCV to the double-acting hydraulic cylinder.

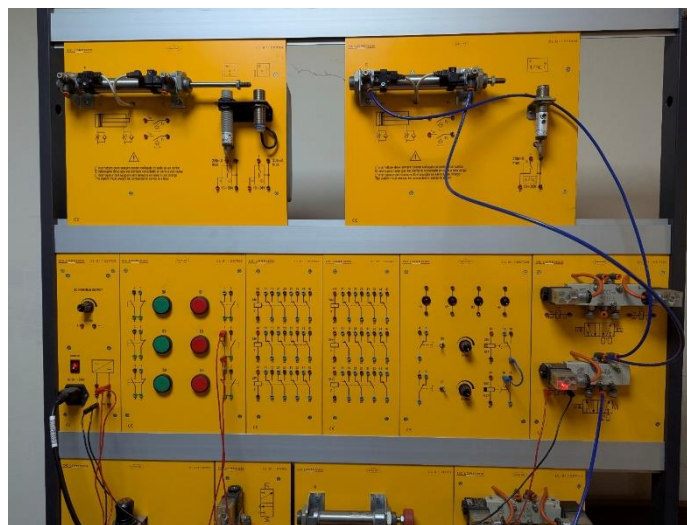


Fig.5.2. Circuit Buildup for Unstable Electro-drive with Unstable Control ^[2]

Condition 02: Unstable Electro-drive with Stable Control

Electro-drive: A push button is used, creating an unstable electro-drive as it returns to the off position when released.

Control: Switches are installed on both sides of the directional control valve (DCV), ensuring the valve maintains a steady position after actuation.

Setup:

- Connect the push button to the solenoid valve.
- Set up the DCV with switches on both sides.
- Link the DCV to the double-acting cylinder.

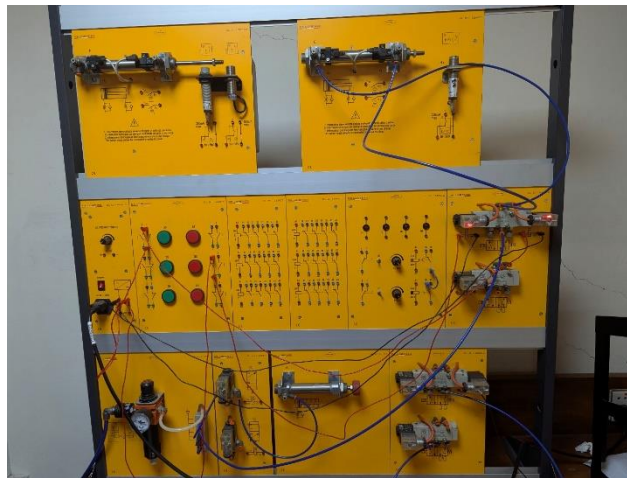


Fig.5.3. Circuit Buildup for Unstable Electro-drive with Stable Control ^[2]

Condition 03: Stable Electro-drive with Unstable Control

Electro-drive: A stable electro-drive is achieved by using a constant power source or a latching mechanism to keep the solenoid activated.

Control: A spring return on one side of the directional control valve (DCV) causes unstable control by forcing the valve to reset automatically.

Setup:

- Connect a stable power source or latching mechanism to the solenoid valve.
- Configure the DCV with a spring return on one side.
- Link the DCV to the double-acting cylinder.

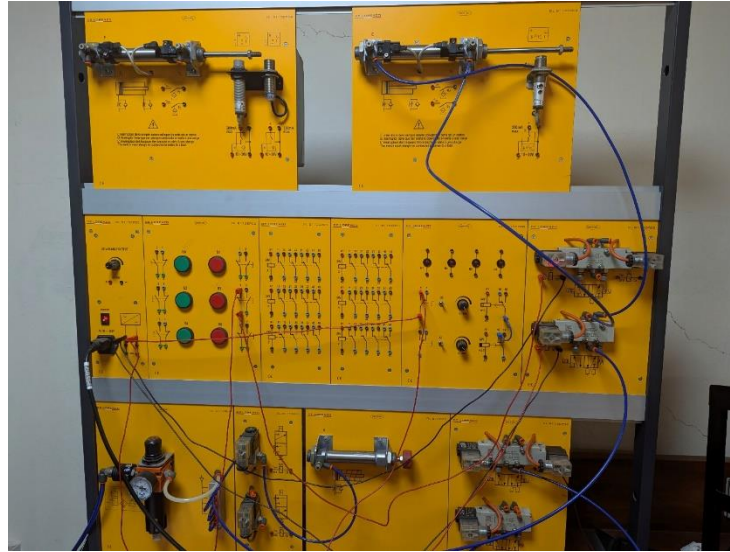


Fig.5.4. Circuit Buildup for Stable Electro-drive with Unstable Control ^[2]

Condition 04: Stable Electro-drive with Stable Control

Electro-drive: A stable electro-drive is ensured by using a continuous power source or a latching mechanism to maintain solenoid activation.

Control: Switches installed on both sides of the directional control valve (DCV) keep the valve in a stable and consistent position.

Setup:

- Connect a stable power source or latching mechanism to the solenoid valve.
- Configure the DCV with switches on both sides.
- Link the DCV to the double-acting cylinder.



Fig.5.5. Circuit Buildup for Stable Electro-drive with Stable Control ^[2]

Applications

Electro-pneumatic systems using double-acting cylinders are extensively applied in industrial automation, where stable control is crucial for dependable operation. Major applications include:

1. Packaging Machinery: Power sealing, cutting, and pressing actions, requiring consistent operation to maintain product quality.
2. Robotics: Drive robotic grippers and actuators, where stability ensures accurate pick-and-place tasks.
3. Automotive Manufacturing: Support assembly and testing operations where precision and safety are essential.

Discussion

This experiment evaluated the behavior of a double-acting cylinder under four different conditions. In Condition 01 (unstable electro-drive and control), cylinder motion was inconsistent due to unreliable actuation and valve resetting. Condition 02 (unstable electro-drive with stable control) showed better movement, though minor inconsistencies remained. In Condition 03 (stable electro-drive with unstable control), erratic behavior persisted as the spring return disrupted valve operation. Finally, Condition 04 (stable electro-drive and control) achieved smooth, predictable cylinder motion. The results confirmed that consistent electro-drive and valve control are vital for reliable electro-pneumatic system performance, matching theoretical expectations.

Conclusion

The experiment successfully demonstrated the impact of stable and unstable controls on a double-acting cylinder's performance. Smooth and predictable operation, as seen in Condition 04, required both stable electro-drive and control. In contrast, Conditions 01 and 03 showed erratic movement under unstable setups, while Condition 02 offered partial improvement. These findings emphasize the importance of consistent actuation and valve control for designing reliable electro-pneumatic systems in industrial automation.

References

- [1] VComp Inc, "What are single acting & double acting hydraulic cylinders," *Medium*, 18-Dec-2017. [Online]. Available: <https://medium.com/@vcompmississauga/what-are-single-acting-double-acting-hydraulic-cylinders-817d6e6f30a8>. [Accessed: 17-Apr-2025].
- [2] "Basic modular electro-pneumatics trainer," *De Lorenzo Training & Didactical Equipment*, 10-Apr-2024. [Online]. Available: <https://delorenzoglobal.com/en/product/basic-electropneumatics-trainer/>. [Accessed: 17-Apr-2025].